

HUANCHENG CHEN

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EDUCATION

The University of Texas at Austin, Austin, Texas 2020 – 2025

Ph.D. in Electrical and Computer Engineering

South China University of Technology, Guangzhou, China 2015 – 2019

B.E. in Electrical Engineering and Automation

RESEARCH OVERVIEW

My research interests lie in LLM post-training and efficient LLMs, with a focus on making large language models more capable, reliable, and deployable. Specifically, I am interested in:

- **LLM post-training**: SFT, RL for agents to learn memory management and self-evolving skills. [13, 14]
- **Efficient LLMs**: inference acceleration via quantization, pruning, and distillation. [3, 4, 9, 12]
- **Continual learning**: fine-tuning pretrained LLMs and VLMs on downstream tasks. [8, 9, 10, 11]
- **Trustworthy AI**: privacy-preserving machine learning and model safety. [2, 5]

INDUSTRIAL EXPERIENCE

Microsoft AI, Mountain View, California Mar. 2026 – present

Applied Scientist II

- Leading architecture design (profile modeling, memory management), LLM training (data curation, SFT/RL), and inference optimization for **Copilot Search** AI mode enabling conversational and agentic search.
- Pretraining/post-training multimodal retrieval model on PB-scale text, image, and video data for **Bing**.
- Building token-efficient, low-latency and resilient AI grounding platform **Web IQ** (beta testing).

Accenture, Mountain View, California Jun. 2025 – Mar. 2026

Senior Research Scientist at Advanced AI Center

- Contributed to Accenture's research on memory management for long-horizon LLM agents via **MemexRL**.
- Contributed to Accenture's research on efficient RL for improving SLM's tool-use capabilities via **CacheRL**.
- Contributed to the Training-as-a-Service functionality (training, evaluation, and data curation), an end-to-end pipeline that enables clients to customize LLMs with their own data on the **AI Refinery** platform.

Sony AI, Tokyo, Japan May. 2024 – Aug. 2024

Research Intern at PPML Team

- Designed **MAC**, a zero-shot method that enhances spatial semantics in diffusion-based layout-to-image generation without additional training.

Sony AI, Austin, Texas Feb. 2024 – May. 2024

Research Intern at PPML Team

- Proposed **DualLoRA**, a novel PEFT method for continually fine-tuning pretrained ViTs on downstream tasks without data replay.

Toyota InfoTech Lab, Mountain View, California May. 2022 – Aug. 2022

Research Intern at Infrastructure & Data Platform

- Applied knowledge distillation in **FedHKD** to mitigate the model drift problem in non-IID distributed learning and protect local data privacy.

- Developed a U-2-Net-based framework for removing irrelevant background to improve flaw detection.

PUBLICATIONS

(★ indicates equal contribution)

- [TMLR] [10] S. Cha★, **H. Chen**★, H. Vikalo, “FedProTIP: Task-Agnostic Federated Continual Learning via Replay-Free Gradient Projection”, *Transactions on Machine Learning Research*, 2026.
- [ICLR] [9] D. Kim, S. Cha, **H. Chen**, C. Wang, H. Vikalo, “Quantized Gradient Projection for Memory-Efficient Continual Learning”, *International Conference on Learning Representations*, 2026.
- [CVPRW] [8] **H. Chen**, J. Li, W. Zhuang, C. Chen, L. Lyu, “Replay-Free Continual Low-Rank Adaptation with Dynamic Memory”, *Computer Vision and Pattern Recognition Workshop*, 2026.
- [CVPRW] [7] **H. Chen**, J. Li, W. Zhuang, H. Vikalo, L. Lyu, “Training-Free Layout-to-Image Generation with Marginal Attention Constraints”, *Computer Vision and Pattern Recognition Workshop*, 2026.
- [NeurIPS] [6] **H. Chen**, H. Vikalo, “Heterogeneity-Guided Client Sampling: Towards Fast and Efficient Non-IID Federated Learning”, *Advances in Neural Information Processing Systems*, 2024.
- [ICML] [5] **H. Chen**, H. Vikalo, “Recovering Labels from Local Updates in Federated Learning”, *International Conference on Machine Learning*, 2024.
- [CVPR] [4] **H. Chen**, H. Vikalo, “Mixed-Precision Quantization for Federated Learning on Resource-Constrained Heterogeneous Devices”, *Computer Vision and Pattern Recognition*, 2024.
- [ICLR] [3] **H. Chen**, C. Wang, H. Vikalo, “The Best of Both Worlds: Accurate Global and Personalized Models through Federated Learning with Data-Free Hyper-Knowledge Distillation”, *International Conference on Learning Representations*, 2023.
- [CVPRW] [2] **H. Chen**, H. Vikalo, “Federated Learning in Non-IID Settings Aided by Differentially Private Synthetic Data”, *Computer Vision and Pattern Recognition Workshop*, 2023. **Oral**
- [ICCVW] [1] A. Mohamed★, **H. Chen**★, Z. Wang, C. Claudel, “Skeleton-Graph: Long-Term 3D Motion Prediction From 2D Observations Using Deep Spatio-Temporal Graph CNNs”, *International Conference on Computer Vision Workshop*, 2021.

PREPRINTS

- [14] M. Islam, S. Thakur, **H. Chen**, S. Park, J. Wang, G. Kim, “CacheRL: Multi-Turn Tool-Calling Agents via Cached Rollouts and Hybrid Reward”, *Technical Report*.
- [13] Z. Wang, **H. Chen**, J. Wang, W. Wei, “Memex(RL): Scaling Long-Horizon LLM Agents via Indexed Experience Memory”, *Technical Report*.
- [12] S. Cha, **H. Chen**, D. Kim, H. Zhang, K. Chan, G. Veciana, H. Vikalo, “CoreQ: Learning-Free Mismatch Correction and Successive Rounding for Quantization”, *Under Review*.
- [11] D. Kim, A. Wynter, **H. Chen**, H. Kim, H. Vikalo, “Foundation-Preserving Adaptation via Generalized Rayleigh-Quotient Optimization”, *Under Review*.

SKILLS

- Platforms: Linux, Azure, AWS, GCP
- Programming: Python, MATLAB, Java, C/C++, $\LaTeX$
- Agent & training stacks: PyTorch, verl, slime, SGLang, vLLM, Ray, SkyPilot, Kubernetes

COMMUNITY SERVICES

- Conference Reviewer:
ICML (22,23,24,25,26), NeurIPS (22,23,24,25,26), ICLR (24,25), IJCAI (24,25), AAAI (25), CVPR (24,25), ACL (26).
- Journal Reviewer:
IEEE Transactions on Mobile Computing